Uday Kumar Guduguntla

AI Engineer | GenAI & NLP | Python

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Professional Summary

- AI Engineer with **2+ years** of Python experience, including **1+ year** building Generative AI (GenAI) and NLP solutions: **Retrieval-Augmented Generation (RAG)** systems, chatbots and **multi-agent** applications on a multi-tenant SaaS platform. Skilled in LLMs, LangChain, tool calling and FastAPI.

Skills

Programming - Python, SQL

GenAI & NLP - Generative AI (GenAI), Natural Language Processing (NLP), LLMs, spaCy, NLTK, RAG, Embeddings, Agentic AI, Multi-Agent Systems, Prompt Engineering, Context Engineering, Tool Calling, MCP, Synthetic Data Generation

Frameworks & APIs - LangChain, FastAPI, Pydantic, REST APIs

Vector Databases & Search - Chroma DB, FAISS, Pinecone, Meilisearch, Semantic Search

Databases & Graphs - PostgreSQL, MySQL, Redis, ClickHouse, Neo4j, Cypher, Ontology Design

Tools & Streaming - Git, GitHub, Docker, Kafka, Debezium

Experience

Trainee Software Engineer - Python **Oct 2025 – Present**

Worktual Innovations Pvt. Ltd. *Chennai, Tamil Nadu*

- Develop GenAI and NLP solutions for a multi-tenant SaaS platform spanning contact center, CRM, ticketing and campaign products, partnering with product teams to turn business needs into features.
- Architected the multi-tenant **RAG system**, the **Change Data Capture (CDC)** pipeline from PostgreSQL to Neo4j and the **Cypher agent** for the cross-product analytics platform.
- Delivered multi-channel chatbots that use business knowledge and tools to book appointments and raise tickets.
- Collaborated with teammates on an in-house Python agent framework, ticketing agent improvements and synthetic data pipelines for LLM fine-tuning and product testing, including about **26,000** conversation samples.

Python Developer - Backend & Tools **Jun 2024 – Sep 2025**

Webfosys Networks Pvt. Ltd. *Remote / Dubai, UAE*

- Automated publishing across multiple WordPress sites with a **FastAPI** system for metadata, images and scheduling.
- Engineered an article quality checker that combines rule-based SEO checks with **LLM**-based scoring, using score thresholds to approve articles or flag low-scoring ones for revision.
- Developed a Python tool using the Google Search Console API to track queries, clicks, CTR and page performance.
- Contributed to the ZenX partnership, where Webfosys and ZenX teamed up to build a cross-border digital payments app, helping define payment workflows and backend requirements.

Projects

Enterprise Knowledge Retrieval Engine (RAG) | *Worktual*

- Built an ingestion pipeline that detects file type, extracts text from PDF, DOCX, TXT and Excel files and splits the text into retrieval-ready chunks.
- Organized chunks into tenant-level collections that keep each organization's data isolated, currently serving **12–14 tenants**.
- Developed Meilisearch-based retrieval APIs with metadata filters and faceted search, delivering **10–20 ms** retrieval.
- Supplied the retrieved context to chatbots and LLM agents to ground answers in business knowledge.

Omnichannel Conversational AI Assistants | *Worktual*

- Developed custom GenAI chatbots for websites, Facebook and Instagram for healthcare, jewellery and fitness clients.
- Wrote system prompts and business rules for each client to control agent behavior and responsibilities.
- Grounded answers in each client's knowledge base using the RAG system.
- Integrated tool calling for lead capture, appointment booking and support tickets, so agents carry out real operations instead of only describing actions.
- Implemented intent-based routing and Sales-to-Service agent handoffs that preserve the full chat context.

Cross-Product Analytics Copilot | *Worktual*

- Co-designed a three-layer multi-agent platform (Super Agent, Domain Agents, SQL and Cypher executors) that answers business questions across products and about **800 database tables**.
- Engineered the Cypher agent independently to write and run Neo4j queries and choose the next relationship to explore for related data and possible causes.
- Added rolling context summarization to keep long graph investigations within LLM context limits.

Instagram AI Chatbot (LangChain) | *Personal Project*

- Developed an Instagram DM chatbot that generates LLM replies with **LangChain**, using a FastAPI webhook connected to the official Meta Instagram API.
- Used LangChain prompts and chains to turn incoming messages into relevant replies.
- Deployed the webhook service on Render and upgraded the bot from rule-based replies to LLM-generated replies.

Agentic AI Framework (Internal ADK) | *Worktual*

- Co-built a reusable Python framework for agents, teams and multi-level orchestration with context-preserving handoffs.
- Added tool calling with Pydantic input validation, on-demand modular skills and PostgreSQL-backed session memory.
- Integrated an MCP client so agents can use tools from remote MCP servers.

Synthetic Data Engine for LLM Fine-Tuning & Product Testing | *Worktual*

- Designed, with a team of two, a scenario-based pipeline using Gemini and NVIDIA APIs that generated about **26,000** conversation samples with tool calls for LLM fine-tuning, checked by an LLM-as-a-judge layer and human review.
- Contributed to a schema-driven generator that creates relational test data with correct primary, foreign and cross-schema keys and validates the output, to test whether products handle reports and data at production level.

Skill-Based Ticketing Agent | *Worktual*

- Took ownership of an existing ticketing agent: removed duplicate tools, added missing ones and restructured prompts with a colleague.
- Added modular skills and moved critical steps into deterministic backend tools, reducing context size and improving reliability.

Real-Time Knowledge Graph Sync Engine (CDC) | *Worktual*

- Independently built a Python CDC pipeline using PostgreSQL, Debezium and Kafka that keeps a separate Neo4j graph up to date for each organization through YAML mappings.
- Implemented tenant routing, idempotent graph updates, backfill and a dead-letter queue (DLQ) for failed events, and contributed to the enterprise ontology.

Education

Master of Computer Applications (MCA) **2023 – 2025**

Narayana Engineering College, Nellore *CGPA: 8.8*

- Top 40 finalist among 4,000+ participants in the CVCORP Hackathon.
- Won 1st prize in the DevFista Hackathon, organized by Nexzera and SkillBridge.
- Won 1st prize in Code Change at TechChroma, a national-level technical symposium.

B.Sc. Statistics **2020 – 2023**

Vivekananda Degree College, Nellore *CGPA: 9.2*

Extracurricular Activities

- **Technical Trainer, CodeTantra:** Trained **1,000+** students in Python, C and Data Structures.
- Led trainer teams at SRM (Chennai), VEMU (Chittoor) and SSEPTP (Putaparthi); covered absent teammates' classes.
- **Event Leadership:** Ran a full-stack web workshop for 70+ students and led TechChroma, a national symposium.